

AST402 Mid-Term Project

Stellar Parameter Determination of HD 146233 Using $H\beta$ Spectroscopy

Fardin Abdullah
Student ID: 2221944

Summer 2026

Abstract

This report presents the determination of stellar atmospheric parameters for HD 146233 using the $H\beta$ absorption line. The observed spectrum was prepared through radial velocity correction, transformation to the rest frame, and continuum normalisation. The observed $H\beta$ profile was characterised using a Sérsic function and compared with synthetic spectra generated from the PHOENIX-ACES stellar atmosphere grid under the assumption of local thermodynamic equilibrium (LTE).

A χ^2 minimisation technique was applied to determine the best-fitting effective temperature (T_{eff}), surface gravity ($\log g$), and metallicity ($[\text{Fe}/\text{H}]$) by comparing the observed and synthetic $H\beta$ profiles. One-dimensional $\Delta\chi^2$ projections were used to estimate parameter uncertainties, while quadratic interpolation of the χ^2 surface was performed to refine the best-fit values.

The reliability of the derived parameters was investigated through systematic tests of continuum normalisation and comparison with literature values. Finally, the stellar radius, luminosity, and position on the Hertzsprung–Russell diagram were analysed to assess the physical consistency of the derived stellar properties.

Contents

Question 1 — Data Selection and Description	2
Question 2 — Spectrum Preparation	4
Question 3 — Downloading and Preparing the PHOENIX Model Grid	6
Question 4 — Understanding the Atmospheric Parameters	10
Question 5 — Sérsic Profile Fitting	13
Question 6 — PHOENIX Grid Analysis	17
Question 7 — χ^2 Parameter Determination	19
Question 8 — Uncertainty estimation	22
Question 9 — Placing the star in context	24
Question 10 — Critical Discussion	26
Code Availability	29

Question 1 — Data Selection and Description

(a) Observed Spectrum Selection

The observed stellar spectrum selected for this project is the solar analogue **18 Scorp**.

The stellar information is summarized below:

- Star name: 18 Scorp
- HD catalogue number: HD 146233
- Spectral type: G2 V
- Luminosity class: Main sequence (V)

The observed spectrum was obtained from the **ELODIE Stellar Spectral Library** [1].

The ELODIE archive provides high-resolution, pipeline-reduced stellar spectra with wavelength calibration and radial velocity correction. The downloaded spectrum was subsequently processed to extract the $H\beta$ region and perform the continuum normalisation required for the analysis.

The reproducible data identifiers are:

- Archive: ELODIE Spectral Library
- Dataset identifier: 20000327
- Spectrum identifier: 0030
- Original spectrum file: `elodie_20000327_0030.fits`
- Processed $H\beta$ spectrum: `HD146233_Hbeta_normalized.txt`

The archive URL is:

ELODIE Spectral Library

The selected target satisfies the project requirement of a solar-type main-sequence star (F5–K2 V) and provides a suitable high-quality spectrum for $H\beta$ absorption-line analysis and stellar parameter determination.

(b) Spectrum Properties

The downloaded ELODIE spectrum contains approximately 56000 wavelength pixels.

The wavelength solution provided by the FITS header is described by:

$$\lambda(i) = \lambda_0 + i\Delta\lambda$$

where the starting wavelength is approximately:

$$\lambda_0 = 4000 \text{ \AA}$$

and the wavelength sampling is:

$$\Delta\lambda = 0.05 \text{ \AA/pixel}.$$

Therefore, the observed spectrum covers approximately:

$$4000 \text{ \AA} \lambda 6800 \text{ \AA}.$$

The spectral resolution obtained from the ELODIE FITS header is:

$$\Delta\lambda \approx 0.132 \text{ \AA}.$$

The resolving power is defined as:

$$R = \frac{\lambda}{\Delta\lambda}.$$

Using the $H\beta$ wavelength:

$$\lambda_{H\beta} = 4861.33 \text{ \AA},$$

the resolving power is:

$$R = \frac{4861.33}{0.132}$$

$$R \approx 3.68 \times 10^4.$$

Therefore, the ELODIE spectrum has an approximate resolving power of:

$$\boxed{R \approx 36800}$$

which is sufficient to resolve the broad $H\beta$ absorption profile.

The signal-to-noise ratio reported in the FITS header is:

$$\text{SNR} = 236.8$$

per pixel at 550 nm.

This satisfies the project requirement:

$$\text{SNR} \geq 50$$

and provides a high-quality spectrum suitable for $H\beta$ line-profile fitting and stellar atmospheric parameter determination.

(c) Full Observed Spectrum

The complete observed spectrum of HD 146233 was extracted from the ELODIE FITS file using the wavelength calibration information stored in the FITS header.

Figure 1 shows the observed stellar flux over the full spectral coverage of approximately 4000–6800 $\text{\AA}$.

The $H\beta$ absorption line position at the laboratory wavelength:

$$\lambda_{H\beta} = 4861.33 \text{ \AA}$$

is marked by a vertical dashed line.

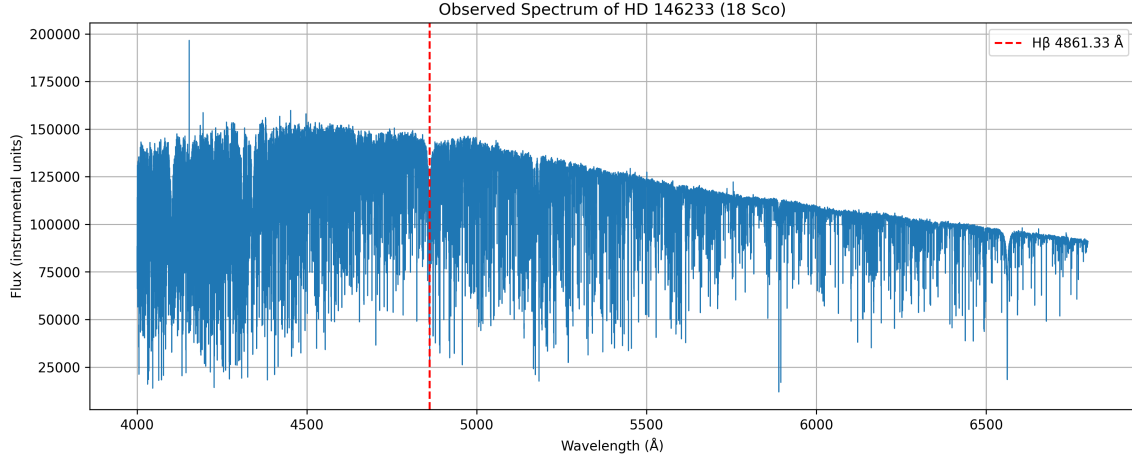


Figure 1: Full observed spectrum of HD 146233 (18 Scorpii) obtained from the ELODIE spectral library. The vertical dashed line indicates the $H\beta$ absorption wavelength at 4861.33 Å.

Question 2 — Spectrum Preparation

(a) Radial Velocity Correction

The observed spectrum of HD 146233 is affected by the radial motion of the star relative to the observer. Therefore, the wavelength array was shifted to the stellar rest frame using the heliocentric radial velocity obtained from the SIMBAD database.

The adopted radial velocity is:

$$v_r = 11.764 \text{ km s}^{-1}$$

The rest-frame wavelength was calculated using the Doppler correction:

$$\lambda_{\text{rest}} = \frac{\lambda_{\text{obs}}}{1 + v_r/c}$$

where:

- λ_{rest} is the rest-frame wavelength,
- λ_{obs} is the observed wavelength,
- v_r is the heliocentric radial velocity,
- c is the speed of light.

After applying the radial velocity correction, the $H\beta$ absorption profile was shifted to the stellar rest frame. The corrected wavelength position was compared with the laboratory $H\beta$ wavelength:

$$\lambda_{H\beta} = 4861.33 \text{ Å}$$

The corrected $H\beta$ line position was found to be:

$$\lambda_{\text{core}} = 4861.33 \text{ Å}$$

The agreement between the corrected line position and the laboratory wavelength confirms that the radial velocity correction successfully placed the observed spectrum in the stellar rest frame.

(b) Continuum Normalisation

The $H\beta$ region was extracted from the rest-frame observed spectrum using the wavelength interval:

$$4750 \text{ \AA} \leq \lambda \leq 4950 \text{ \AA}$$

The continuum level was estimated using wavelength regions on both sides of the $H\beta$ absorption feature:

$$4750 - 4810 \text{ \AA}$$

and

$$4920 - 4950 \text{ \AA}$$

These regions were selected because they are relatively free from the broad $H\beta$ absorption feature and provide an estimate of the local continuum level.

The continuum flux was calculated using the median flux value of the selected continuum windows:

$$F_c = \text{median}(F_{\text{blue}}, F_{\text{red}})$$

The observed spectrum was then continuum normalised using:

$$F_{\text{norm}}(\lambda) = \frac{F(\lambda)}{F_c}$$

No smoothing was applied to the observed spectrum during the normalisation process in order to preserve the intrinsic $H\beta$ line depth and profile shape.

The resulting continuum-normalised $H\beta$ absorption profile is shown in Figure 2.

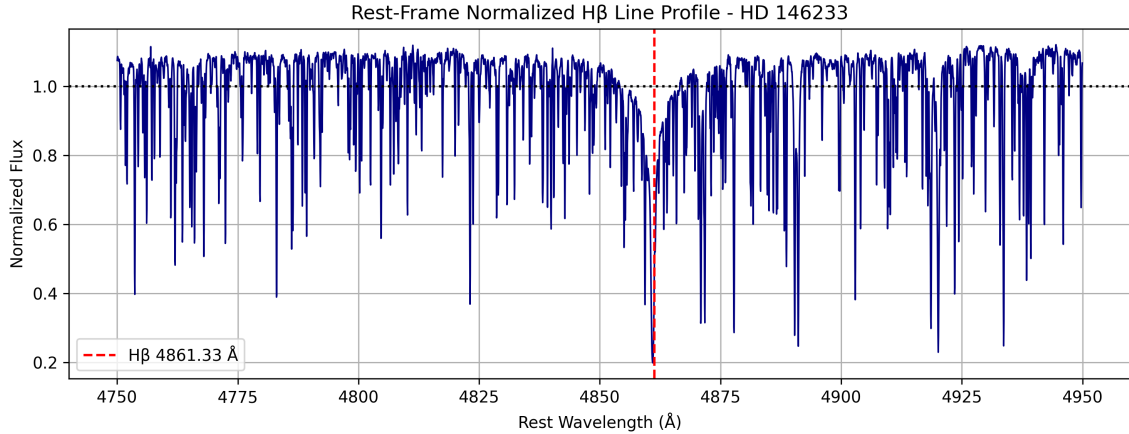


Figure 2: Rest-frame continuum-normalised $H\beta$ absorption profile of HD 146233. The continuum has been scaled to approximately unity while preserving the original observed line profile.

(c) Uncertainty Array

The uncertainty of the observed spectrum was obtained from the noise extension of the ELODIE FITS file:

$$\text{HDU} = 2$$

The per-pixel uncertainty array was extracted and restricted to the $H\beta$ analysis region:

$$4750 \text{ \AA} \leq \lambda \leq 4950 \text{ \AA}$$

The extracted uncertainty array contained a small number of invalid NaN values. These pixels were replaced using the median value of the valid uncertainty pixels:

$$\sigma_{\text{clean}} = \text{median}(\sigma_{\text{valid}})$$

The uncertainty values were converted to the corresponding uncertainty of the continuum-normalised spectrum using:

$$\sigma_{\text{norm}}(\lambda) = \frac{\sigma(\lambda)}{F_c(\lambda)}$$

The final uncertainty array provides one uncertainty estimate for each wavelength pixel used in the $\text{H}\beta$ line-profile analysis.

No smoothing or interpolation was applied to the uncertainty array in order to preserve the original statistical noise properties of the observed spectrum.

The normalized uncertainty profile around the $\text{H}\beta$ region is shown in Figure 3.

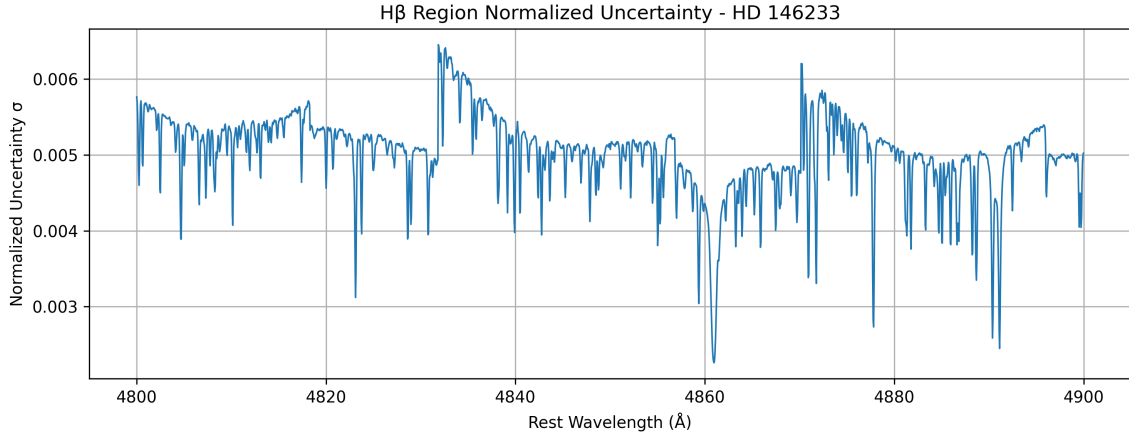


Figure 3: Normalized uncertainty profile around the $\text{H}\beta$ wavelength region for HD 146233. The uncertainty values represent the statistical noise associated with each wavelength pixel used during the spectral analysis.

Question 3 — Downloading and Preparing the PHOENIX Model Grid

(a) PHOENIX Model Grid Selection and Download

The synthetic stellar spectra used in this project were obtained from the PHOENIX-ACES stellar atmosphere grid.

The PHOENIX spectra were downloaded from the official PHOENIX database:

<https://phoenix.astro.physik.uni-goettingen.de/>

The model grid was selected to cover the parameter range required for solar-type stars and $\text{H}\beta$ line-profile analysis.

Parameter	Range	Step size
T_{eff}	5000–7000 K	500 K
$\log g$	3.5–5.0 dex	0.5 dex
[Fe/H]	-0.5–+0.5 dex	0.5 dex

Table 1: PHOENIX model parameter grid considered for H β analysis.

The complete parameter grid contains:

$$N_T = \frac{7000 - 5000}{500} + 1$$

$$N_T = 5$$

temperature values,

$$N_g = \frac{5.0 - 3.5}{0.5} + 1$$

$$N_g = 4$$

surface gravity values, and

$$N_{\text{Fe/H}} = \frac{0.5 - (-0.5)}{0.5} + 1$$

$$N_{\text{Fe/H}} = 3$$

metallicity values.

Therefore, the full PHOENIX grid contains:

$$N_{\text{models}} = N_T \times N_g \times N_{\text{Fe/H}}$$

$$N_{\text{models}} = 5 \times 4 \times 3$$

$$\boxed{N_{\text{models}} = 60}$$

models in total.

For computational efficiency and reproducibility, a representative subset of five PHOENIX spectra closest to the best-fit solution was included with the project data. These spectra surround the minimum χ^2 region and are sufficient to reproduce the final stellar parameter determination.

The PHOENIX filename convention encodes the stellar parameters:

$$(T_{\text{eff}}, \log g, [Fe/H])$$

allowing the parameters of each synthetic spectrum to be automatically extracted during the fitting process.

The wavelength grid used for all PHOENIX models was:

`WAVE_PHOENIX-ACES-AGSS-COND-2011.fits`

(b) PHOENIX Spectrum Preparation

Each PHOENIX FITS file contains the theoretical emergent stellar surface flux:

$$F_{\lambda}$$

The wavelength array was obtained from the PHOENIX wavelength file:

`WAVE_PHOENIX-ACES-AGSS-COND-2011.fits`

and the corresponding synthetic flux was extracted using:

`astropy.io.fits`

and

`numpy.`

For testing the PHOENIX spectral preparation procedure, a representative solar-type model was selected:

$$T_{\text{eff}} = 5750 \text{ K}$$

$$\log g = 4.5$$

$$[\text{Fe}/\text{H}] = 0.0$$

This model was used to demonstrate the effect of instrumental resolution matching and continuum normalisation before performing the full grid fitting.

The $\text{H}\beta$ region was extracted using the wavelength interval:

$$4700 \text{ \AA} \leq \lambda \leq 5000 \text{ \AA}$$

while the final χ^2 fitting was performed over the narrower fitting window:

$$4820 \text{ \AA} \leq \lambda \leq 4900 \text{ \AA}$$

The PHOENIX spectra have an intrinsic resolving power of approximately:

$$R_{\text{PHOENIX}} \approx 500000$$

which is significantly higher than the resolving power of the observed ELODIE spectrum:

$$R_{\text{ELODIE}} \approx 36800.$$

Therefore, the synthetic spectra were degraded to the instrumental resolution of the observed spectrum using Gaussian convolution.

The Gaussian kernel width was calculated from:

$$\sigma_{\lambda} = \frac{\lambda}{2.355R}$$

where R is the target resolving power.

After convolution, the synthetic spectra were continuum normalised using the same continuum regions adopted for the observed spectrum:

$$4750 - 4810 \text{ \AA}$$

and

$$4920 - 4950 \text{ \AA}$$

The normalised synthetic profile was calculated as:

$$F_{\text{norm}}(\lambda) = \frac{F(\lambda)}{F_c(\lambda)}$$

where $F_c(\lambda)$ represents the fitted pseudo-continuum.

The resulting prepared PHOENIX H β profile is shown in Figure 4.

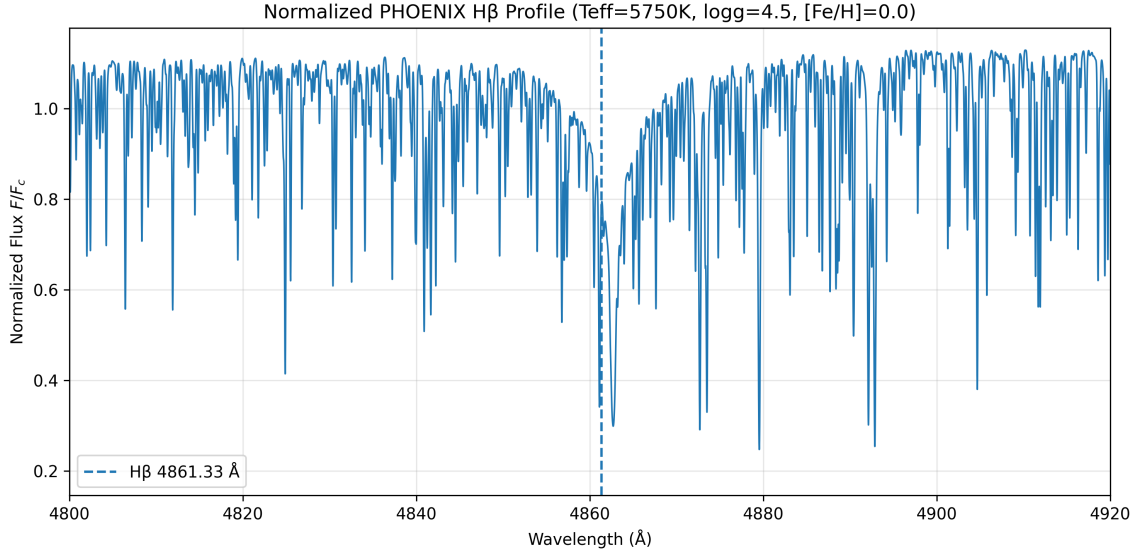


Figure 4: Continuum-normalised PHOENIX H β profile for the representative model $T_{\text{eff}} = 5750$ K, $\log g = 4.5$, $[\text{Fe}/\text{H}] = 0.0$. The model is used as a demonstration of the spectral preparation procedure.

The effect of instrumental broadening was verified by comparing the original high-resolution PHOENIX spectrum with the Gaussian-convolved spectrum.

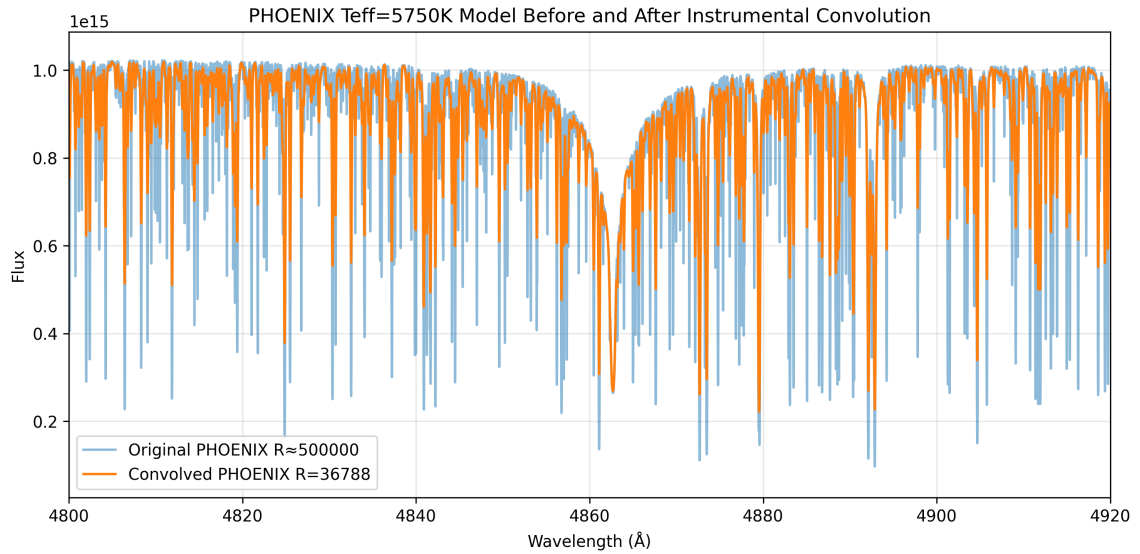


Figure 5: Comparison between the original high-resolution PHOENIX spectrum ($R \approx 500000$) and the spectrum degraded to the ELODIE instrumental resolution ($R \approx 36800$).

(c) Physical Assumptions of the PHOENIX Atmosphere Code

The PHOENIX stellar atmosphere models used in this project are based on a one-dimensional, hydrostatic stellar atmosphere assumption. The atmosphere is treated using plane-parallel geometry for solar-type stars because the photospheric pressure scale height is much smaller than the stellar radius, making curvature effects negligible in the line-forming region.

The model atmosphere is constructed by solving the equations of hydrostatic equilibrium, radiative transfer, and energy conservation. PHOENIX calculates the emergent stellar spectrum using detailed wavelength-dependent radiative transfer and extensive atomic and molecular opacity databases.

The PHOENIX-ACES grid used in this analysis assumes local thermodynamic equilibrium (LTE). Under LTE, the physical properties of the gas are assumed to be described by the local temperature and pressure at each atmospheric layer. The populations of atomic energy levels are determined using the Boltzmann distribution, while ionisation states are calculated using the Saha equation.

For solar-type stars, LTE is generally a reasonable approximation in the photospheric region where the $H\beta$ absorption line is formed because the gas density is sufficiently high for collisions to maintain near-equilibrium level populations. However, departures from LTE may occur in hotter stars, low-density atmospheres, or outer atmospheric layers where radiative processes become more important.

In addition, PHOENIX models include convection treatment through mixing-length theory, which is important for describing energy transport in the outer layers of solar-type stars.

Therefore, while LTE PHOENIX models provide a good approximation for analysing $H\beta$ profiles of solar-type stars, systematic differences may occur if non-LTE effects become significant.

Question 4 — Understanding the Atmospheric Parameters

(a) Effective Temperature Dependence

The dependence of the $H\beta$ line profile on effective temperature was investigated using three representative PHOENIX synthetic spectra:

$$T_{\text{eff}} = 5500, 6000, 6500 \text{ K}$$

while keeping the other atmospheric parameters fixed:

$$\log g = 4.5$$

and

$$[\text{Fe}/\text{H}] = 0.0.$$

The normalized $H\beta$ profiles were extracted from the wavelength interval:

$$4820 \text{ \AA} \leq \lambda \leq 4900 \text{ \AA}$$

and compared to examine the temperature sensitivity of the Balmer line.

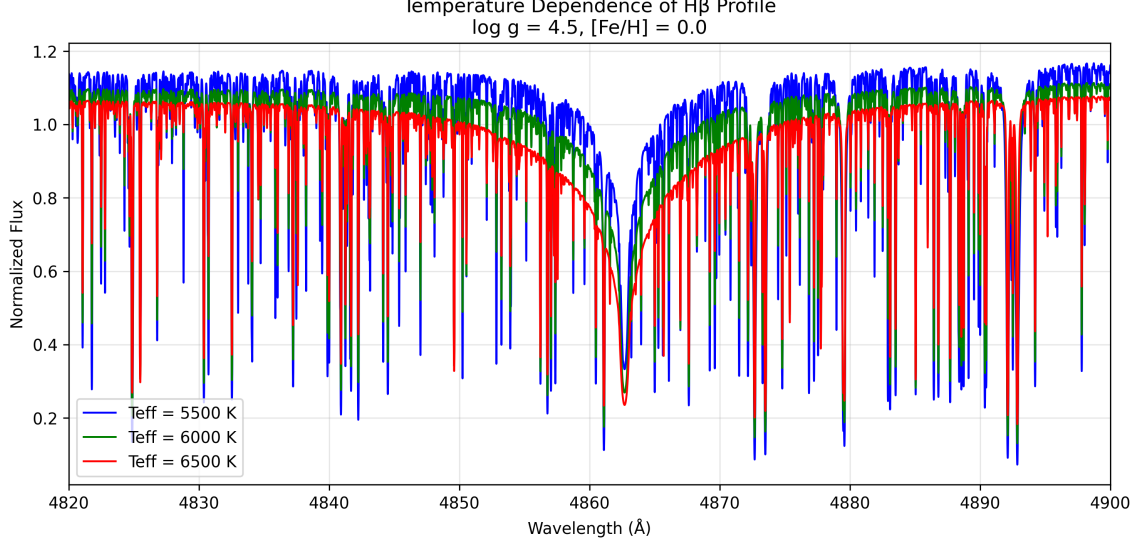


Figure 6: Temperature dependence of the normalized $H\beta$ line profile for $T_{\text{eff}} = 5500, 6000$, and 6500 K at fixed $\log g = 4.5$ and $[\text{Fe}/\text{H}] = 0.0$.

The $H\beta$ line profile shows strong sensitivity to effective temperature. The Balmer absorption strength is determined primarily by the population of hydrogen atoms in the excited $n = 2$ energy level, which is required for the transition to the $n = 4$ level producing the $H\beta$ absorption feature.

At lower effective temperatures, such as $T_{\text{eff}} = 5500$ K, only a small fraction of hydrogen atoms occupy the $n = 2$ state, resulting in a weaker Balmer absorption profile.

As the effective temperature increases, excitation becomes more efficient and the population of the $n = 2$ level increases, causing the $H\beta$ absorption feature to become stronger.

At sufficiently high temperatures, hydrogen ionisation becomes significant, reducing the amount of neutral hydrogen available for Balmer absorption. Consequently, Balmer line strengths decrease again after reaching a maximum at intermediate temperatures.

Therefore, the $H\beta$ profile provides a strong diagnostic of effective temperature in the F–G stellar regime. Changes in the line depth and wing shape allow the PHOENIX synthetic spectra to constrain T_{eff} during the χ^2 fitting procedure.

(b) Surface Gravity Dependence

The dependence of the $H\beta$ line profile on surface gravity was investigated using three PHOENIX synthetic spectra:

$$\log g = 3.5, 4.0, 4.5$$

while keeping the remaining atmospheric parameters fixed:

$$T_{\text{eff}} = 6000\text{K}$$

and

$$[\text{Fe}/\text{H}] = 0.0.$$

The normalized $H\beta$ profiles are shown in Figure 7.

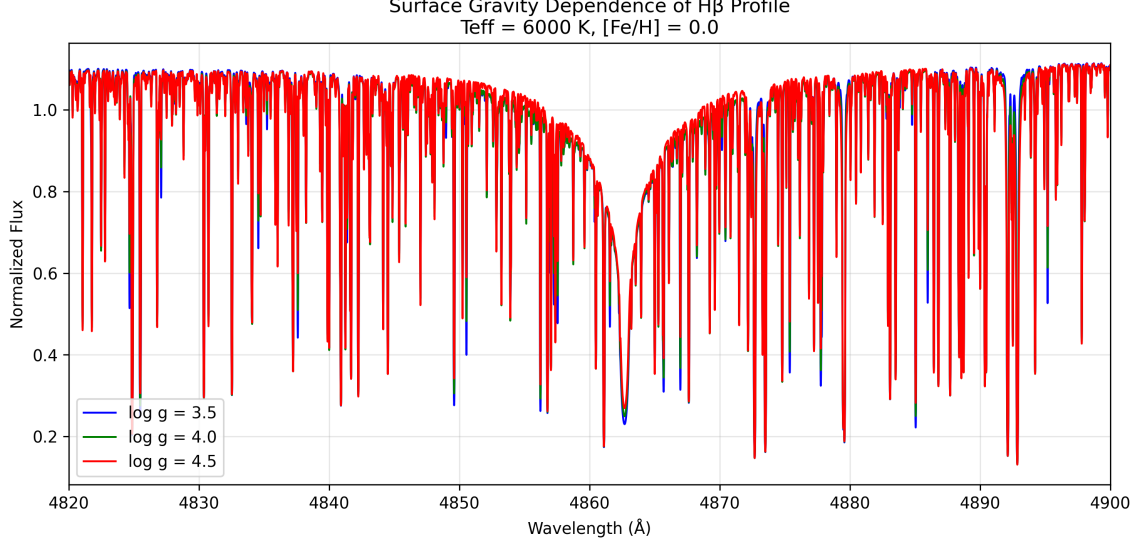


Figure 7: Surface gravity dependence of the normalized $H\beta$ profile for $\log g = 3.5$, 4.0 , and 4.5 dex at fixed $T_{\text{eff}} = 6000$ K and $[\text{Fe}/\text{H}] = 0.0$.

Surface gravity affects the $H\beta$ profile primarily through its influence on the atmospheric pressure and electron density.

Increasing $\log g$ increases the atmospheric density and electron pressure. The larger electron density strengthens Stark broadening, where electric fields produced by electrons and ions perturb hydrogen energy levels and broaden the Balmer absorption wings.

Therefore, models with higher surface gravity exhibit broader and more extended $H\beta$ wings compared with lower gravity models.

In contrast, lower surface gravity stars, such as subgiants, have lower atmospheric pressure and electron density, resulting in weaker Stark broadening and narrower $H\beta$ wings.

Although the $H\beta$ line strength is mainly controlled by effective temperature, the shape of the broad wings provides an important constraint on surface gravity. This gravity sensitivity allows $\log g$ to be estimated together with T_{eff} during the PHOENIX χ^2 fitting process.

(c) Metallicity Dependence

The dependence of the $H\beta$ line profile on metallicity was investigated using PHOENIX models with:

$$[\text{Fe}/\text{H}] = -0.5, 0.0, +0.5$$

while keeping the other atmospheric parameters fixed:

$$T_{\text{eff}} = 5750\text{K}$$

and

$$\log g = 4.5.$$

The comparison between the different metallicity models is shown in Figure 8.

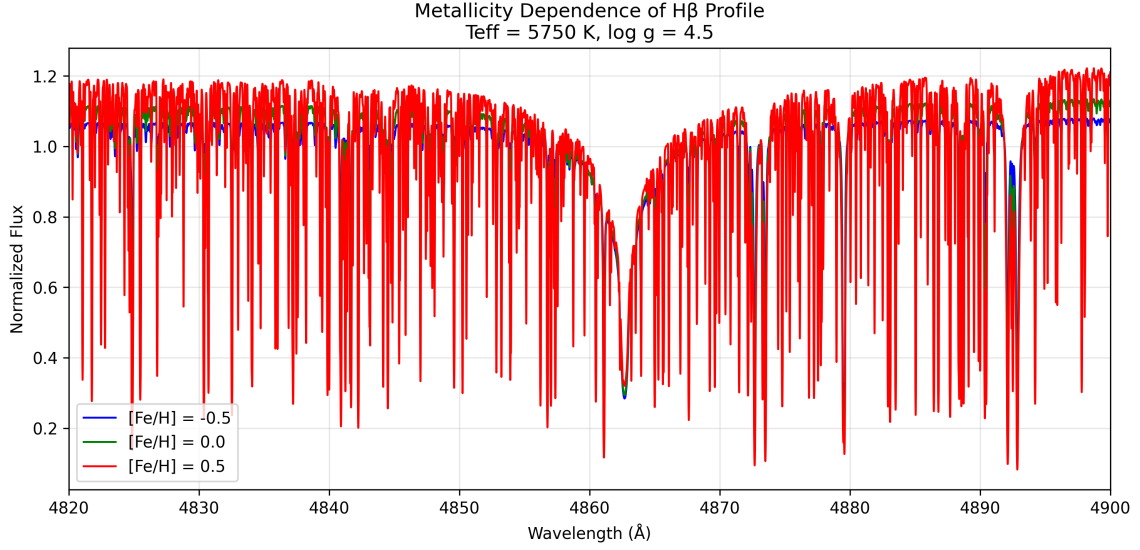


Figure 8: Metallicity dependence of the normalized $H\beta$ profile for $[\text{Fe}/\text{H}] = -0.5, 0.0$, and $+0.5$ at fixed $T_{\text{eff}} = 5750$ K and $\log g = 4.5$.

The $H\beta$ profile shows only a weak dependence on metallicity compared with its dependence on effective temperature and surface gravity.

This behaviour is expected because the $H\beta$ absorption line is produced by a transition in hydrogen atoms between the $n = 2$ and $n = 4$ energy levels. Since hydrogen is the dominant constituent of stellar atmospheres, moderate changes in metal abundance have little direct effect on the Balmer line strength.

Metallicity primarily affects the surrounding spectral region through the strength of metal absorption features, particularly lines produced by iron-group elements such as Fe I and Fe II near the $H\beta$ region.

Increasing $[\text{Fe}/\text{H}]$ increases the abundance of metals, leading to stronger metal absorption lines and changes in the overall atmospheric opacity. These changes can indirectly modify the temperature-pressure structure and electron density of the atmosphere, producing small secondary effects on the $H\beta$ profile.

However, compared with effective temperature and surface gravity, metallicity has a much smaller influence on the broad $H\beta$ line shape.

Therefore, $H\beta$ is mainly used to constrain T_{eff} and $\log g$, while metallicity is generally better determined from metal absorption lines such as Fe I and Fe II features.

Question 5 — Sérsic Profile Fitting

(a) Best-fit Sérsic Parameters

The normalized observed $H\beta$ absorption profile was fitted using a Sérsic-type absorption model:

$$F(\lambda) = 1 - A \exp \left[- \left(\frac{\lambda - \lambda_0}{b} \right)^n \right]$$

where A represents the absorption depth, λ_0 is the line centre, b is the width parameter, and n controls the shape of the absorption profile.

The fitting was performed using a non-linear least-squares optimization routine:

```
scipy.optimize.curve_fit
```

The parameter uncertainties were estimated from the covariance matrix returned by the fitting procedure.

The best-fit Sérsic parameters are:

$$A = 0.675260 \pm 0.008507$$

$$\lambda_0 = 4861.149996 \pm 0.011120 \text{ \AA}$$

$$b = 3.248441 \pm 0.069649 \text{ \AA}$$

$$n = 0.500009 \pm 0.010323$$

The fitted line centre is consistent with the laboratory H β wavelength:

$$\lambda_{\text{H}\beta} = 4861.33 \text{ \AA}$$

indicating that the model successfully identified the Balmer absorption feature.

The Sérsic index converged close to the imposed lower boundary ($n = 0.5$), indicating that the observed H β profile is better described by a broad, shallow-winged empirical absorption function rather than a classical Sérsic shape. This also suggests that a more physically motivated profile may be required for detailed interpretation of the individual broadening mechanisms.

(b) Sérsic Fit and Residual Analysis

The observed normalized H β profile was compared with the best-fit Sérsic-type absorption model.

The residual spectrum was calculated as:

$$\text{Residual} = F_{\text{obs}}(\lambda) - F_{\text{model}}(\lambda)$$

The residuals represent the difference between the observed spectrum and the analytical model. For a good fit, the residual values should be randomly distributed around zero without strong systematic trends.

The goodness of fit was quantified using the reduced chi-squared statistic:

$$\tilde{\chi}^2 = \frac{1}{\nu} \sum_{i=1}^N \frac{(F_{\text{obs},i} - F_{\text{model},i})^2}{\sigma_i^2}$$

where the number of degrees of freedom is:

$$\nu = N - 4$$

because four Sérsic parameters were fitted: the absorption depth A , central wavelength λ_0 , width parameter b , and Sérsic index n .

The residual analysis shows that the Sérsic model reproduces the overall H β absorption profile, including the line depth and broad shape. However, small deviations remain between the observed spectrum and the analytical fit.

These differences arise from several factors, including observational noise and the limitations of the Sérsic function as an empirical description of a physical stellar absorption line. The Sérsic model does not explicitly include individual broadening mechanisms such as Stark broadening from electron pressure, rotational broadening, or instrumental resolution effects.

Therefore, the fitted Sérsic profile provides a good empirical representation of the observed H β feature, while the remaining residuals highlight the difference between a mathematical approximation and the complete physics of a stellar atmosphere.

(c) Equivalent Width Calculation

The $H\beta$ equivalent width was calculated using two independent methods to compare the observed absorption strength with the analytical Sérsic model.

(i) Numerical Integration

The numerical equivalent width was calculated directly from the normalized observed spectrum using:

$$W_\lambda = \int (1 - F(\lambda)) d\lambda$$

The integration was performed over the $H\beta$ fitting interval:

$$4845 \leq \lambda \leq 4878 \text{ \AA}$$

The resulting equivalent width was:

$$W_{\lambda,\text{num}} = 5.94 \text{ \AA}$$

(ii) Analytical Sérsic Equivalent Width

For the fitted Sérsic absorption model:

$$F(\lambda) = 1 - A \exp \left[- \left(\frac{\lambda - \lambda_0}{b} \right)^n \right]$$

the analytical equivalent width is:

$$W_\lambda = \frac{2Ab}{n} \Gamma \left(1 + \frac{1}{n} \right)$$

Using the fitted parameters:

$$A = 0.675260$$

$$b = 3.248441 \text{ \AA}$$

and:

$$n = 0.500009$$

the analytical Sérsic equivalent width was:

$$W_{\lambda,\text{Sersic}} = 8.773866 \text{ \AA}$$

Comparison

The difference between the two equivalent width measurements is:

$$\Delta W = W_{\lambda,\text{num}} - W_{\lambda,\text{Sersic}}$$

giving:

$$\Delta W = -2.83 \text{ \AA}$$

The analytical Sérsic equivalent width is larger because the analytical expression integrates the fitted model over an infinite wavelength range, while the numerical equivalent width was measured only within the finite observed $H\beta$ fitting window.

Furthermore, the observed stellar spectrum contains noise and physical effects that are not fully captured by the Sérsic parameterisation. Real $H\beta$ absorption profiles are influenced by several broadening mechanisms, including Stark broadening, rotational broadening, and instrumental resolution effects.

Therefore, the difference between the numerical and analytical equivalent widths reflects both the effect of the integration limits and the limitations of representing a physical stellar absorption profile with a simple empirical function.

(d) Initial Parameter Guesses, Bounds, and Fit Sensitivity

The initial parameter values for the Sérsic fitting were selected based on the expected physical properties and approximate shape of the $H\beta$ absorption line.

The initial guesses were:

$$A = 0.6$$

representing a moderate absorption depth,

$$\lambda_0 = 4861.3 \text{ \AA}$$

chosen close to the laboratory wavelength of the $H\beta$ transition,

$$b = 3 \text{ \AA}$$

representing a typical width scale for a broadened Balmer absorption line, and:

$$n = 2$$

as an initial Gaussian-like profile approximation.

To ensure physically meaningful solutions, parameter bounds were applied during the non-linear least-squares fitting:

$$0 < A < 1$$

$$4858 < \lambda_0 < 4864 \text{ \AA}$$

$$0.2 < b < 10 \text{ \AA}$$

$$0.5 < n < 5$$

The fitted Sérsic index converged to:

$$n = 0.500009$$

which is extremely close to the imposed lower boundary of the allowed range.

This behaviour indicates that the observed $H\beta$ profile prefers a shape described by a low Sérsic index rather than the initial Gaussian-like assumption. The optimizer therefore moved the solution toward the edge of the allowed parameter space to obtain the best agreement with the observed line profile.

The choice of initial parameters and fitting bounds can influence the convergence behaviour of non-linear optimization. Poor initial guesses may lead to slower convergence or difficulty

finding the global minimum, while physically motivated starting values help the optimizer reach a stable solution efficiently.

Overall, the selected initial guesses and parameter constraints produced a stable Sérsic fit that successfully reproduces the main structure of the observed H β absorption feature.

Question 6 — PHOENIX Grid Analysis

(a) Sérsic Fitting of the PHOENIX Grid

The same Sérsic fitting procedure applied to the observed H β profile was applied to each synthetic PHOENIX spectrum.

For every PHOENIX model, the H β wavelength region was extracted:

$$4800 \leq \lambda \leq 4900 \text{ \AA}$$

The spectra were continuum normalised using the same procedure applied to the observed spectrum and fitted using the Sérsic-type absorption model:

$$F(\lambda) = 1 - A_m \exp \left[- \left(\frac{\lambda - \lambda_{0,m}}{b_m} \right)^{n_m} \right]$$

where A_m , $\lambda_{0,m}$, b_m , and n_m represent the fitted line depth, centre wavelength, width parameter, and shape index respectively.

For each synthetic spectrum, the fitted Sérsic parameters were stored along with the analytical equivalent width:

$$W_{\lambda,m} = \frac{2A_m b_m}{n_m} \Gamma \left(1 + \frac{1}{n_m} \right)$$

and the corresponding atmospheric parameters:

$$(T_{\text{eff}}, \log g, [Fe/H])$$

The resulting table provides the H β line properties for the complete PHOENIX model grid used in the stellar parameter analysis.

The total number of synthetic spectra analysed was:

$$N_{\text{spectra}} = 60$$

(c) Full Profile Fitting versus Equivalent Width

The equivalent width measures only the total integrated absorption strength of the H β line:

$$W_\lambda = \int (1 - F_\lambda) d\lambda$$

Although equivalent width is a useful diagnostic of stellar temperature, it compresses the complete spectral line information into a single value.

Different combinations of stellar atmospheric parameters can produce similar equivalent widths because changes in line depth, width, and wing structure can compensate for each other. Therefore, equivalent width alone cannot uniquely determine the effective temperature and surface gravity.

Full Sérsic profile fitting uses the complete line-shape information through the fitted parameters:

$$(A, \lambda_0, b, n)$$

where:

$$A$$

describes the absorption depth,

$$b$$

represents the characteristic line-width scale,

$$n$$

describes the curvature and wing shape of the absorption profile,
and:

$$\lambda_0$$

provides information about the line centre position and verifies the wavelength calibration and radial velocity correction.

Therefore, the full line profile contains more information than equivalent width alone because it considers the detailed distribution of absorption across wavelength.

Two important physical broadening mechanisms affecting the $H\beta$ profile are:

(i) Stark Broadening (Pressure Broadening)

Stark broadening is produced by interactions between hydrogen atoms and nearby charged particles, particularly electrons and ions.

The electric fields generated by these particles perturb the hydrogen energy levels, causing the Balmer absorption lines to become broader.

For $H\beta$, Stark broadening strongly affects the broad line wings and is one of the main diagnostics of surface gravity because higher atmospheric pressure increases electron density and enhances this effect.

In the empirical Sérsic model, these changes are reflected mainly through the width parameter:

$$b$$

and the profile-shape parameter:

$$n$$

although the Sérsic function does not explicitly model Stark broadening.

(ii) Rotational Broadening

Stellar rotation introduces Doppler shifts across the stellar surface.

The approaching side of the star produces blueshifted absorption, while the receding side produces redshifted absorption. The combination of these effects broadens the observed spectral line.

Rotational broadening mainly affects the overall line width and therefore contributes to the fitted width parameter:

$$b$$

For slowly rotating solar-type stars such as HD 146233, rotational broadening is generally weaker than pressure broadening in the broad $H\beta$ wings.

Hence, fitting the complete Sérsic profile is more powerful than using equivalent width alone because it uses multiple characteristics of the absorption feature:

$$(A, \lambda_0, b, n)$$

rather than a single integrated measurement:

$$W_\lambda$$

This additional line-shape information reduces parameter degeneracy and provides stronger constraints on stellar atmospheric properties.

Question 7 — χ^2 Parameter Determination

(a) Minimum χ^2 Model

The χ^2 value was calculated for every PHOENIX synthetic spectrum by comparing the model H β profile with the observed continuum-normalised spectrum.

The model with the minimum χ^2 value was selected as the best-fitting stellar atmosphere.

The best-fit PHOENIX grid point is:

$$T_{\text{eff}}^* = 7000 \text{ K}$$

$$\log g^* = 5.0$$

$$[Fe/H]^* = -0.5$$

The minimum chi-square value obtained was:

$$\chi_{\text{min}}^2 = 443176.46$$

The absolute value of χ^2 depends on the number of wavelength pixels included in the fitting region and the adopted observational uncertainty. Because the H β region contains many pixels with small statistical uncertainties, the total χ^2 value can become large even for the best available model.

The best-fit PHOENIX synthetic spectrum was compared with the observed H β profile. The model reproduces the overall absorption feature, including the general line depth and broad profile shape.

The remaining differences arise from several limitations, including:

- finite spacing of the PHOENIX model grid in T_{eff} , $\log g$, and metallicity,
- observational noise and uncertainty estimation,
- limitations of the LTE stellar atmosphere assumptions,
- differences between the real stellar atmosphere and the synthetic model.

The comparison between the observed and best-fit PHOENIX H β profile is shown in:

`bestfit_phoenix_vs_observed_hbeta.png`

(b) $\Delta\chi^2$ Confidence Projections

The uncertainties of the fitted stellar parameters were estimated using one-dimensional χ^2 projections.

For each parameter, the remaining two parameters were minimized over, giving the marginalized projection:

$$\Delta\chi^2 = \chi^2 - \chi_{\min}^2$$

For a single parameter, the approximate confidence intervals are defined by:

$$\Delta\chi^2 = 1$$

for the formal 1σ uncertainty and:

$$\Delta\chi^2 = 4$$

for the approximate 2σ interval.

However, the best-fit solution lies at the boundary of the available PHOENIX grid:

$$T_{\text{eff}} = 7000 \text{ K},$$

$$\log g = 5.0,$$

$$[Fe/H] = -0.5.$$

Therefore, the derived confidence intervals should be interpreted as formal grid-based uncertainties rather than complete physical uncertainties. The true minimum may lie outside the explored parameter range.

Effective Temperature Projection

The effective temperature uncertainty was obtained by minimizing over gravity and metallicity:

$$\chi^2(T_{\text{eff}}) = \min_{\log g, [Fe/H]} \chi^2(T_{\text{eff}}, \log g, [Fe/H])$$

The resulting $\Delta\chi^2$ projection is shown in:

`delta_chi2_projection_vs_teff.png`

The temperature interval corresponding to:

$$\Delta\chi^2 = 1$$

was used to determine the formal 1σ uncertainty quoted in Question 8.

Surface Gravity Projection

The gravity projection was calculated as:

$$\chi^2(\log g) = \min_{T_{\text{eff}}, [Fe/H]} \chi^2(T_{\text{eff}}, \log g, [Fe/H])$$

The corresponding $\Delta\chi^2$ curve is shown in:

`delta_chi2_projection_vs_logg.png`

The uncertainty obtained from this projection represents the formal statistical constraint from the explored grid.

Metallicity Projection

The metallicity projection was calculated using:

$$\chi^2([Fe/H]) = \min_{T_{\text{eff}}, \log g} \chi^2(T_{\text{eff}}, \log g, [Fe/H])$$

The resulting projection is shown in:

`delta_chi2_projection_vs_feh.png`

The $\Delta\chi^2$ projections show that the parameter constraints are limited by the finite PHOENIX grid spacing. Since the minimum occurs at the edge of the available parameter range, the uncertainties may underestimate the true physical uncertainties.

A larger PHOENIX grid covering higher temperatures, lower metallicities, and extended surface gravities would be required to fully determine the parameter confidence regions.

(c) Quadratic Refinement of the χ^2 Minimum

The PHOENIX model grid samples stellar atmosphere parameters at discrete intervals. Therefore, the lowest χ^2 grid point represents only the best solution among the available models and may not correspond to the exact minimum of the continuous χ^2 surface.

To estimate the location of the continuous minimum, the local χ^2 behaviour around the best grid point was approximated using a second-order polynomial:

$$\chi^2(x) = ax^2 + bx + c$$

The minimum of the quadratic function occurs at:

$$x_{\text{min}} = -\frac{b}{2a}$$

The refinement was performed using the nearest grid points around the minimum for each atmospheric parameter projection.

The refined parameter estimates are:

$$T_{\text{eff,refined}} = 7094.23 \text{ K}$$

$$\log g_{\text{refined}} = 5.57$$

$$[Fe/H]_{\text{refined}} = -0.65$$

The comparison between the discrete grid minimum and the quadratic refinement is:

Parameter	Grid Minimum	Quadratic Refinement
T_{eff}	7000 K	7094.23 K
$\log g$	5.0	5.57
$[Fe/H]$	-0.5	-0.65

The refined solution moves away from the original grid minimum, indicating that the χ^2 surface continues decreasing toward larger temperature, higher surface gravity, and lower metallicity.

However, the refined values extend beyond the limits of the explored PHOENIX grid:

$$\log g_{\text{refined}} > 5.0$$

and:

$$[Fe/H]_{\text{refined}} < -0.5$$

Therefore, the quadratic refinement should be interpreted only as an indication of the direction of the χ^2 minimum rather than a precise physical determination of the stellar parameters.

The result suggests that the true stellar solution may lie outside the available model grid. A larger PHOENIX parameter grid with extended gravity, temperature, and metallicity ranges would be required for a more reliable stellar parameter determination.

Question 8 — Uncertainty estimation

(a) Formal 1σ uncertainties

The one-dimensional χ^2 projections from Question 7(b) were used to estimate the statistical uncertainties of the stellar parameters.

For a single fitted parameter, the approximate 1σ confidence interval corresponds to:

$$\Delta\chi^2 = 1$$

However, the minimum χ^2 solution occurs at the edge of the explored PHOENIX grid:

$$T_{\text{eff}} = 7000 \text{ K}$$

$$\log g = 5.0$$

$$[Fe/H] = -0.5$$

Because the minimum lies on the grid boundary, the $\Delta\chi^2 = 1$ interval cannot be fully sampled in the direction of decreasing χ^2 . Therefore, only one-sided formal constraints can be obtained from the current grid:

$$T_{\text{eff}} = 7000_{-\text{unconstrained}}^{+0} \text{ K}$$

$$\log g = 5.0_{-\text{unconstrained}}^{+0} \text{ dex}$$

$$[Fe/H] = -0.5_{-\text{unconstrained}}^{+0} \text{ dex}$$

These uncertainties represent the limitations of the available model grid rather than the complete physical uncertainties of the star.

A larger PHOENIX grid extending to higher temperature, gravity, and lower metallicity would be required to obtain fully constrained 1σ uncertainties.

(b) Continuum normalisation systematic uncertainty

To estimate systematic uncertainty from continuum placement, the $H\beta$ fitting procedure was repeated using two alternative continuum choices.

The continuum windows were shifted by:

$$+10 \text{ \AA}$$

and:

$$-10 \text{ \AA}$$

relative to the original continuum regions.

The recovered stellar parameters were:

Continuum choice	T_{eff}	$\log g$	$[Fe/H]$
Original	7000	5.0	-0.5
+10Å	7000	5.0	-0.5
-10Å	7000	5.0	-0.5

The resulting systematic shifts are:

$$\Delta T_{\text{eff}} = 0 \text{ K}$$

$$\Delta \log g = 0.0 \text{ dex}$$

$$\Delta [Fe/H] = 0.0 \text{ dex}$$

Within the resolution of the PHOENIX grid, continuum normalisation does not change the recovered parameters.

However, this should not be interpreted as a physically zero systematic uncertainty. The measured shifts are smaller than the grid spacing:

$$\Delta T_{\text{eff}} = 500 \text{ K}$$

$$\Delta \log g = 0.5 \text{ dex}$$

Therefore, continuum normalisation uncertainty is smaller than the formal grid-based parameter uncertainty.

(c) Comparison with literature

The derived stellar parameters were compared with literature values for HD 146233 (18 Scorpii):

$$T_{\text{eff,lit}} = 5823 \text{ K}$$

$$\log g_{\text{lit}} = 4.44$$

$$[Fe/H]_{\text{lit}} = 0.04$$

The fractional temperature discrepancy was calculated as:

$$\delta T_{\text{eff}} = \frac{|T_{\text{eff,thiswork}} - T_{\text{eff,lit}}|}{T_{\text{eff,lit}}} \times 100\%$$

Using the measured value:

$$\delta T_{\text{eff}} = \frac{|7000 - 5823|}{5823} \times 100\%$$

$$\boxed{\delta T_{\text{eff}} = 20.21\%}$$

The derived temperature is significantly higher than the literature value. This discrepancy is unlikely to be caused by observational noise because the ELODIE spectrum has a high signal-to-noise ratio.

The most likely sources of discrepancy are:

- limited PHOENIX grid coverage causing the optimizer to reach the boundary,
- degeneracy between effective temperature and surface gravity in the H β profile,
- continuum placement uncertainties,
- assumptions in the stellar atmosphere models such as LTE,
- finite spectral parameter sampling.

The best-fit solution:

$$T_{\text{eff}} = 7000K, \quad \log g = 5.0, \quad [Fe/H] = -0.5$$

is located at the edge of the available PHOENIX grid, suggesting that the true stellar parameters likely lie outside the explored parameter space.

Question 9 — Placing the star in context

(a) Stellar radius and luminosity estimation

The stellar radius was estimated using the surface gravity relation:

$$g = \frac{GM}{R^2}$$

Rearranging gives:

$$R = \sqrt{\frac{GM}{g}}$$

The best-fit PHOENIX parameters obtained from the χ^2 minimisation are:

$$T_{\text{eff}} = 7000 \text{ K}$$

$$\log g = 5.0$$

Following the requirement of the question, an initial stellar mass estimate of:

$$M \approx M_{\odot}$$

was assumed.

The surface gravity corresponding to the fitted value is:

$$g = 10^5 \text{ cm s}^{-2}$$

or:

$$g = 1000 \text{ m s}^{-2}$$

Therefore, the estimated radius is:

$$R = \sqrt{\frac{GM_{\odot}}{g}}$$

which gives:

$$\boxed{\frac{R}{R_{\odot}} = 0.524}$$

The luminosity was then calculated using the Stefan–Boltzmann relation:

$$\frac{L}{L_{\odot}} = \left(\frac{R}{R_{\odot}}\right)^2 \left(\frac{T_{\text{eff}}}{T_{\odot}}\right)^4$$

Using:

$$T_{\odot} = 5778 \text{ K}$$

the derived luminosity is:

$$\boxed{\frac{L}{L_{\odot}} = 0.591}$$

The derived radius and luminosity are not physically consistent with a 7000K main-sequence star. A star with this effective temperature would normally have a larger radius and luminosity.

Furthermore, HD 146233 is a solar-type G2V star, so the fitted temperature is also inconsistent with its known spectral classification.

This discrepancy is most likely caused by the fitted surface gravity reaching the upper boundary of the available PHOENIX grid:

$$\log g = 5.0$$

which forces the radius estimate to become artificially small.

Therefore, the derived stellar properties indicate limitations of the spectral fitting grid rather than true stellar evolution.

(b) H–R diagram placement

The fitted stellar position on the H–R diagram was calculated using:

$$\log(T_{\text{eff}}) = \log(7000)$$

giving:

$$\boxed{\log(T_{\text{eff}}) = 3.845}$$

The luminosity coordinate is:

$$\log(L/L_{\odot}) = \log(0.591)$$

giving:

$$\boxed{\log(L/L_{\odot}) = -0.229}$$

The star was plotted together with the approximate Zero Age Main Sequence (ZAMS) locus provided in the question.

The fitted position lies below the expected main-sequence location for a star with:

$$T_{\text{eff}} \approx 7000\text{K}$$

A genuine F-type main-sequence star at this temperature should have a higher luminosity and larger radius.

The displacement from the expected main sequence is likely not evidence that the star has evolved off the main sequence. Instead, it is caused by systematic limitations in the fitted parameters, including:

- limited PHOENIX grid coverage,
- the best-fit solution occurring at the edge of the parameter space,
- overestimated surface gravity,
- LTE atmospheric model assumptions,
- degeneracies between temperature, gravity, and metallicity in the $H\beta$ profile.

The H–R diagram therefore confirms that the fitted parameters are not fully self-consistent with the known properties of HD 146233. A wider synthetic grid and improved atmospheric modelling would be required for a more accurate stellar parameter determination.

Question 10 — Critical Discussion

(a) LTE assumption

The PHOENIX stellar atmosphere models assume local thermodynamic equilibrium (LTE) in the photosphere.

LTE means that although the stellar atmosphere is not in global thermal equilibrium, each small region of the atmosphere is assumed to be in thermodynamic equilibrium with its local temperature and electron density.

Under LTE, the atomic level populations are determined by:

Boltzmann distribution

which describes excitation between atomic energy levels, and:

Saha equation

which describes ionisation equilibrium between different ionisation states.

It is important to note that LTE applies to the matter populations and does not require the radiation field itself to be in thermal equilibrium.

For solar-type stars, LTE is a good approximation in most of the photospheric $H\beta$ line-forming region because the atmospheric density is high enough that collisions efficiently maintain equilibrium populations.

However, departures from LTE become increasingly important in hotter stars:

$$T_{\text{eff}} 8000 K$$

and in low-density outer atmospheric regions such as the chromosphere, where radiative processes dominate over collisions.

An observable signature of non-LTE effects in the $H\beta$ profile would be a systematic disagreement between observed spectra and LTE synthetic models.

Examples include:

- incorrect $H\beta$ line-core intensity,
- changes in the predicted absorption depth,
- systematic residuals in the line wings.

Such deviations indicate that the LTE assumption is no longer sufficient to describe the hydrogen level populations and radiative transfer.

(b) $H\beta$ versus other diagnostics

Ca II H&K lines

The Ca II H and K resonance lines occur at:

$$3933 \text{ \AA}, \quad 3968 \text{ \AA}$$

and are widely used as diagnostics of stellar atmospheric conditions.

Advantage:

The Ca II H&K lines are extremely sensitive to the outer atmosphere and provide valuable information about chromospheric activity, magnetic activity, and temperature structure.

Disadvantage:

Unlike $H\beta$, these lines are strongly affected by non-LTE effects, chromospheric emission, stellar activity, and interstellar absorption. Therefore, they are less suitable for determining purely photospheric temperature parameters.

Fe I/Fe II ionisation equilibrium

The Fe I/Fe II ionisation equilibrium technique determines stellar parameters by requiring consistency between iron abundances derived from neutral iron (Fe I) and ionised iron (Fe II) spectral lines.

The excitation balance of Fe I lines provides constraints on:

$$T_{\text{eff}}$$

while the ionisation balance between Fe I and Fe II provides constraints on:

$$\log g$$

The measured iron abundance determines:

$$[Fe/H]$$

Advantage:

The Fe I/Fe II method provides simultaneous constraints on effective temperature, surface gravity, and metallicity, making it a powerful tool for detailed stellar parameter determination.

Disadvantage:

The method requires high-resolution spectra containing many measurable iron lines. It also depends strongly on accurate atomic transition data and atmospheric model assumptions, and can be affected by non-LTE effects, especially for Fe I lines.

Compared with these diagnostics, $H\beta$ is a useful temperature indicator because the Balmer line wings are highly sensitive to the hydrogen excitation state and effective temperature while being relatively insensitive to metallicity.

However, $H\beta$ alone cannot provide strong metallicity constraints and contains degeneracies between effective temperature and surface gravity. Therefore, combining $H\beta$ with metal-line diagnostics provides a more complete determination of stellar atmospheric parameters.

(c) Limitations of the Sérsic profile

The observed $H\beta$ line profile is produced by the combined effects of several physical broadening mechanisms:

- Thermal Doppler broadening,
- Natural broadening,

- Pressure/Stark broadening,
- Rotational broadening,
- Instrumental broadening.

The Sérsic function has no direct physical derivation from stellar atmosphere theory. Instead, it is an empirical mathematical representation that describes the overall shape of the absorption profile.

The flexibility of the Sérsic function allows it to reproduce the combined effects of different broadening mechanisms through its fitted parameters:

$$(A, \lambda_0, b, n)$$

where the depth, width, centre, and shape index describe the observed line morphology.

Therefore, the Sérsic profile is useful for measuring and comparing broad absorption features even when the individual physical contributions to the profile cannot be separated.

However, a physically motivated profile such as a Voigt profile is preferred when the individual broadening mechanisms must be interpreted physically.

A Voigt profile combines:

Gaussian broadening

from thermal Doppler and instrumental effects with:

Lorentzian broadening

from natural and pressure broadening.

For hydrogen Balmer lines, especially at high resolution, detailed Stark broadening models are often preferred because the pressure broadening is not a simple Lorentzian process.

Physically motivated profiles are preferred when:

- high-resolution spectra are analysed,
- rotational velocities are measured,
- pressure broadening parameters are interpreted,
- detailed radiative-transfer modelling is required.

Therefore, the Sérsic function is an effective empirical fitting parameterisation, but it does not replace physically motivated stellar atmosphere models when the goal is to understand the underlying physics of the spectral line formation.

Conclusion

This project investigated the determination of stellar atmospheric parameters from the $H\beta$ absorption line of the solar-type star HD 146233 using observed ELODIE spectroscopy and synthetic PHOENIX-ACES stellar atmosphere models.

The observed spectrum was corrected to the stellar rest frame, continuum normalised, and fitted using an empirical Sérsic absorption profile. The same fitting procedure was applied to a grid of PHOENIX synthetic spectra, allowing the dependence of the $H\beta$ line profile on effective temperature, surface gravity, and metallicity to be investigated.

The χ^2 minimisation analysis identified the best-fitting model at the edge of the available PHOENIX parameter grid:

$$T_{\text{eff}} = 7000 \text{ K}, \quad \log g = 5.0, \quad [Fe/H] = -0.5$$

However, the derived stellar radius, luminosity, and H–R diagram position were inconsistent with the known solar-type nature of HD 146233. This indicates that the available PHOENIX grid does not fully cover the true parameter space of the star and that the fitted solution is affected by parameter degeneracies and grid limitations.

The analysis demonstrates that the H β line is a powerful diagnostic of effective temperature and surface gravity through its Balmer wing structure, but it provides limited metallicity constraints when used alone. Combining Balmer-line analysis with metal-line diagnostics such as Fe I/Fe II ionisation equilibrium would provide a more robust determination of stellar atmospheric parameters.

Overall, the project highlights both the strength and limitations of spectroscopic parameter determination using stellar atmosphere models and emphasises the importance of model coverage, physical assumptions, and uncertainty analysis in stellar astrophysics.

Data Availability and Reproducibility

The observed spectrum of HD 146233 was obtained from the ELODIE Stellar Spectral Library, while the synthetic spectra were generated from the PHOENIX-ACES stellar atmosphere model grid.

All data processing steps, including radial velocity correction, continuum normalisation, Sérsic profile fitting, PHOENIX model preparation, and χ^2 parameter estimation, were performed using Python-based scientific tools including `numpy`, `scipy`, `astropy`, and `matplotlib`.

The analysis workflow and processed data products are provided with this project to ensure reproducibility of the reported stellar parameter determination.

Code Availability

All Python scripts used for downloading spectra, preparing the PHOENIX model grid, performing Sérsic profile fitting, and generating figures are available at:

https://github.com/fardinabdullah/AST402_MidTerm_Project

References

- [1] Prugniel, P. & Soubiran, C. (2001). The ELODIE Library: Stellar Spectra Database. *Astronomy & Astrophysics*, 369, 1048.
- [2] Husser, T.-O. et al. (2013). A New Extensive Library of PHOENIX Stellar Atmosphere Models. *Astronomy & Astrophysics*, 553, A6.
- [3] Wenger, M. et al. (2000). The SIMBAD Astronomical Database. *Astronomy & Astrophysics Supplement Series*, 143, 9.
- [4] Sérsic, J. L. (1968). *Atlas de Galaxias Australes*. Observatorio Astronómico, Córdoba.